

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-9879
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders With Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows how a standard laptop order moves through the ServiceNow platform, from the employee's request to approval, fulfillment and notification. The details of each component and characteristic are given in Table-1 and Table-2.

Architecture: Automated Standard Laptop Ordering on ServiceNow

Scope: Employee ordering, cost-based approval, stock check, task creation, email notification and reporting, all inside ServiceNow.

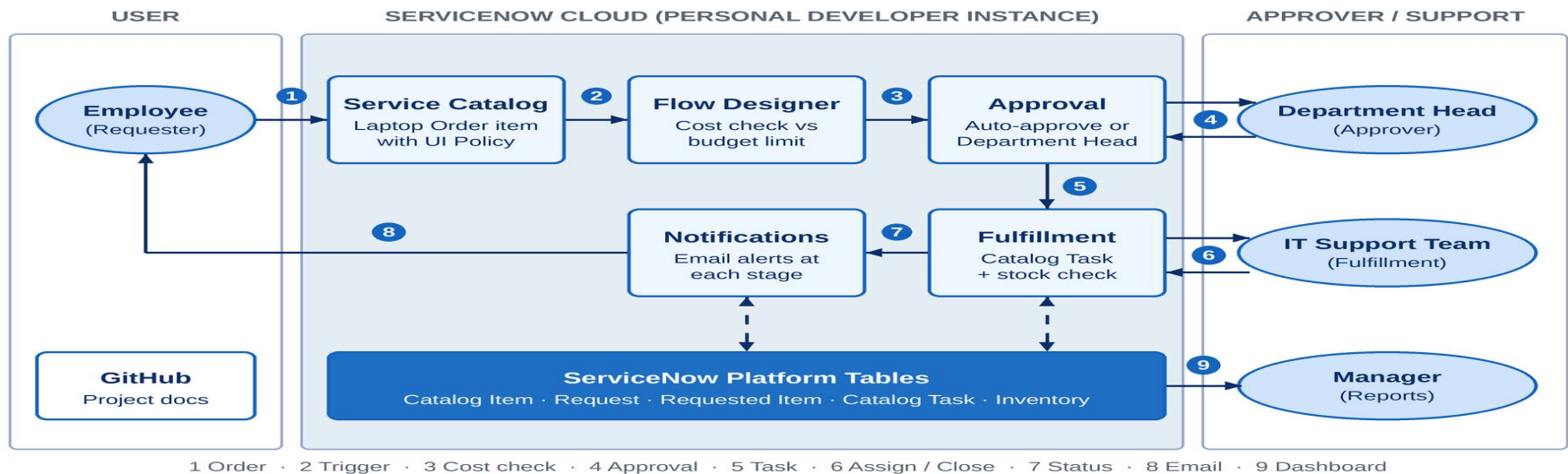


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees place laptop orders through the Service Catalog form, with dynamic RAM and storage options.	ServiceNow Service Catalog, Catalog Item variables, Catalog UI Policy
2.	Application Logic-1	Compares the order cost with the approved budget limit and routes the request: auto-approve or send to the Department Head.	ServiceNow Flow Designer (If condition, Ask for Approval)
3.	Application Logic-2	After approval, creates the fulfillment catalog task for the IT Support group and checks stock in the Inventory table.	Flow Designer Create Catalog Task action, Inventory table lookup
4.	Application Logic-3	Sends stage-wise email notifications (placed, approved / rejected, delivered) to the requester.	ServiceNow Notifications (email)
5.	Database	Stores the catalog item, request, requested item, catalog task and laptop stock records in relational tables.	ServiceNow platform tables (Catalog Item, Request, Requested Item, Catalog Task, Inventory)
6.	Cloud Database	Tables are hosted in the ServiceNow cloud instance, so no separate database service is needed.	ServiceNow cloud (Now Platform database)
7.	File Storage	Project documents and screenshots are kept in a shared repository; request attachments use platform storage.	GitHub; ServiceNow Attachments
8.	External API-1	No external API is required for this project.	None
9.	External API-2	Not applicable; all integrations are native to ServiceNow.	None
10.	Machine Learning Model	Machine learning is not used in this project.	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on a cloud-hosted ServiceNow instance Local Server Configuration: Not required Cloud Server Configuration: ServiceNow Personal Developer Instance (PDI)	ServiceNow Personal Developer Instance (cloud)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework is used; the solution is built with ServiceNow's native low-code features.	ServiceNow Now Platform (low-code)
2.	Security Implementations	Mandatory login; access to requests and tasks is limited by roles (requester, approver, IT support, administrator) and the data stays in the organization's instance.	Role-based access control, ACLs, instance authentication
3.	Scalable Architecture	Every order runs in its own flow execution, so many requests are handled in parallel by the same flow on a multi-tier cloud platform.	Flow Designer on the Now Platform cloud
4.	Availability	The instance is hosted and maintained by ServiceNow, and the flow execution history can be reviewed in Flow Designer if a failure occurs.	ServiceNow cloud hosting, Flow Designer execution history
5.	Performance	Flows run on the server side and the catalog task is created within about 2 minutes of approval; reports and dashboards give a quick view of order status.	Flow Designer server-side execution, Reports and Dashboards